# Customer Relationship Management: Lifecycle of Predicting Customer Lifetime Value

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***Abstract*— In more recent years, customer lifetime value (CLV) has been a popular research topic in the field of customer relationship management. Many models, methods and approaches have been used to predict CLV. However, there are no readily available integrated guidelines for its prediction. This paper proposes an iterative approach for predicting CLV, which highlights the common causes of success and failure in terms of business performance management. An initial computer-assisted template is proposed for CLV prediction and related research.**

***Keywords: customer lifetime value; customer relationship management; iterative approach***

## I. INTRODUCTION

Customer lifetime value (CLV) has been a popular research topic in customer relationship management (CRM) in recent years. It measures what a customer is worth to an organization by referring to the calculation of predicted future transactions and costs [1]. CRM managers often use CLV to define alternative marketing strategies for the organizations, so that the customer service and marketing campaigns can be improved [2]. CLV is also used to drive the business decision making process on how much should a company invest on a customer to yield a higher return on investment (ROI) [3].

Today, organizations tend to steer towards the direction of customer centric strategy. In this regard, customers have become the organizations' number one asset [4]. Therefore, the CLV of a customer could well serve as a good indicator for predicting an organization's incomes from their customers in the future. Moreover, CLV has also been claimed to be the best basis for decisions concerning the market and customer strategy [2].

CLV prediction involves various dynamic impact factors. The predictive inputs from the customers, enterprises and social environmental aspects can be used as input variables to the CLV formula. Many models and methods are available for predicting the CLV. Thus, researchers need some guidelines and a proper plan to kick-start the CLV prediction research.

Although there are many research papers on roadmap and research processes in the business performance management (BPM) domains ([5] – [12]), there are no obvious references for the research of CLV prediction.

This paper makes an initial attempt to propose the lifecycle of CLV prediction research and implementation. It highlights the causes of failure and provides useful tips for resolving problems in the lifecycle.

## II. LITERATURE REVIEW

### A. Customer Relationship Management

CRM is a management field that deals with whatever a company needs to manage and maintain its customer relationships. The activities in CRM include the capture and analysis of customer information, and the execution of business analytics to leverage the output towards better sales performance [13]. It is claimed in [14] that CRM is about introducing the right product or service to the right customer at the right time through the right channel to satisfy the customer's increasing demands. It is further claimed in [14] that the CRM practices can be divided into campaign-centric and customer-centric. However, if one were to focus only on the campaign-centric, one may have neglected the customers' needs and demands. In the end, the customer loyalty may be lost.

In order to succeed in CRM development, four (4) key metrics have been identified to bring values and benefits to a company's CRM policy. These key metrics are CLV, customer retention rate, customer increased response or transaction rate, and return on investment (ROI). Each key metric has its own characteristics and relationship with other key metrics to form a solid bond on the CRM success development. The description of each of these key metrics is shown in TABLE I.

### B. Customer Lifetime Value

Customer lifetime value (CLV) has a long history since the 1930's. Its metrics were originally designed to assess the net present value (NPV) of a customer's future expenses on a company [17]. However, its definition has been re-defined by marketing gurus like Don Peppers and Martha

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Rogers in the 1990s to become their own take of CLV. They introduced more conceptual items into the CLV calculation mix to form a better solution base.

TABLE I. DESCRIPTIONS OF CRM KEY METRICS

| Key Metric | Description |
|---|---|
| CLV | The potential revenue obtained from a customer during his or her relationship with a company [14]. |
| Customer retention rate | The retained earnings divided by total after-tax earnings, expressed as a percentage [15]. |
| Customer increase response / transaction rate | The measurement on the customer response level after making promotions and marketing efforts. Transaction rate refers to the frequency of the customer dealing with the organization over a period of time. A customer's increase response rate may not necessary increase the transaction rate of the customer with the company. |
| Return on Investment (ROI) | The benefit of an investment (gain from investment – cost of investment) is divided by the cost of the investment. The result is expressed as a percentage or a ratio [16]. |

Besides the CLV definition defined by [14] in the previous section, other researchers have also given various definitions to suit their research agenda. Other CLV definitions include:

- The net value of all net payments from the moment the marketing efforts start towards a potential customer, until the customer definitely stops being a customer of the company [2].
- The sum of the cumulated cash flows – discounted using the Weighted Average Cost of Capital (WACC) of a customer over his or her entire lifetime with the company [18].
- The present value of all future profits generated from a customer over his or her life relationship with a company [19].
- The total value of direct contributions and indirect contributions to overhead and profit of an individual customer during the entire customer life cycle, from the start of the relationship to its projected ending [20].

Along with the CLV research, there are several modified versions of CLV formula. All the formulas are derived from the most basic CLV formula as in (1).

CLV = (total customer revenue) * (number of loyal years) * (company profit margin) (1)

The most popular simplified version of the CLV formula is shown in (2). Basically, the popular variables that have been used in the CLV calculation are discounted rates, profit margins, cost and frequency of purchase. Other variables include probabilities, functions, RFM (recency, frequency and monetary), product or service details and customers' personal details. Each variable carries different weight in the calculation formula. Other CLV formulas can be found in ([21] – [28]).

$$\mathrm{CLV} = \sum \frac{NP_t}{(1+r)^t} \quad (2)$$

### C. Importance of Customer Lifetime Value

CLV is applicable to almost every industry. It has gained well acceptance from the company's management to assist their business management process. Similarly, CLV researchers use predictive analytics to boost the output. Some 51.7% of survey respondents among the vendors and consultants in [29] also claimed that CLV modeling was important to their companies and they were willing to invest into it. Svanholmer [2] also shows the importance and wide application of CLV by providing thirteen (13) examples of how managers can use CLV in their decision process.

When CLV is considered in a company's CRM policy, it will ensure that the company gets more details and understanding on the customer behaviors through the CLV analysis process. Torcy [1] also comments that CLV will force the organizations to consider what drives and impedes customer profitability, what are the different types of customer values and what role each should play in determining the organization's strategy and tactics to manage the customer relationship. Companies can maintain high CLV on each of its customers if they know more about the customers and respond promptly to their needs.

## III. CLV PREDICTION LIFECYCLE

### A. Overview

The key factors used in predictive business performance management (BPM) can be applied in the CLV prediction lifecycle. The five key factors are business process, people, technology, prevailing industry standards and internal characteristics of an organization. These key factors are incorporated in the proposed CLV prediction lifecycle as shown in Figure 1. Each phase in the lifecycle is elaborated in the following sections.

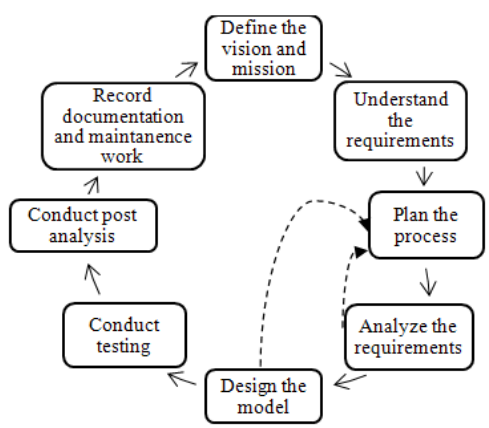


Figure 1. The iterative lifecycle of CLV prediction



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### B. Phase 1: Define the Vision and Mission

The first critical step to success in CLV prediction is to define the vision of the organization. A clear vision specifies the ulterior aims through the appropriate planning and strategy definition. A vision is supported by a set of objectives and goals to make sure that it brings profits at the end of any major engagement.

Many CLV prediction projects have failed because the team defines the vision according to the process but not its ulterior aim. In fact, a vision statement should be established during the initial phase of the lifecycle. Therefore, it is worth investing sufficient time to truly understand an organization's unique ulterior aims before attempting to define the vision, followed by the most suitable mission executable by the organization.

### C. Phase 2: Understand the Requirements

Before understanding the requirements of CLV prediction, it is important to note the reasons leading to the poor performance of predictive models. Firstly, the predictive business performance management projects often used wrong historical data sets which show unclear trends and patterns. In addition, the data sets contained biased information after using the wrong screening and filtering criteria. Some developers also skewed the data into clusters that could not be used to anticipate the future emerging trends and patterns. In general, some of the old predictive models were designed in an overly simplistic way.

The full set of input data is required to conduct proper predictive activities. The requirement settings that need to be identified for CLV prediction are the application domain and setting (contractual or non-contractual), the customer base (individual or group), the application platform (online or offline), the database setting, data collection methods and predictive variables. In all situations, only genuine data should be used. The data range must also be wide enough to enable the predictive analytics to conduct adequate number of experiments. It may be wise to consider using a combination of both historical and real-time input data in order to capture the latest trends and patterns. Besides, it is required to identify what kinds of technology are available in the organization that may be reused in the CLV prediction process, including multi-agent system and business intelligence platforms. Moreover, it is also required to identify the key people who will be working on the CLV prediction lifecycle. The clearer the requirements gathered at the earliest possible stage, the better the planning and execution outcome would be.

### D. Phase 3: Plan the Process

In the planning phase, the established requirements from the previous phase can be used to plan the CLV prediction process. The prediction planning activities include project timelines, milestones, methodologies, structure of the working team, tools, techniques and technology. One common mistake among the unsuccessful predictive model implementation is to select the technology before defining the vision [30]. In fact, the technology must be chosen to fulfill the pre-defined requirements and needs.

As such, it is advisable to use appropriate project management software to record the research project details. Useful features in such software include Gantt chart, human and resource management tool, budgeting and cost management tool, as well as calendaring to record all prediction activities for reference and management purposes. This recording process is useful for future tracking and maintenance jobs.

### E. Phase 4: Analyze the Requirements

Analysis of the gathered requirements is important to ensure that the team members understand what they can gain from the CLV prediction lifecycle. By the end of this phase, the team members should have identified the types of data they need, the methodology they plan to use and the predictive model design they have chosen for the CLV prediction.

The team needs to investigate the predictive variables and predictive models to get the correct variables for CLV prediction. Nineteen (19) dynamic impacts factors have been reviewed in [31]. They are grouped into four categories namely customer, enterprise, customer-enterprise and social related factors. Researchers have been using different variables for CLV prediction under different fields and customer settings ([21] – [28]). As such, a wise approach is to first investigate and analyze the CLV prediction variables across various fields and make comparisons to choose the most suitable set of predictive variables. Furthermore, it is required to analyze the data collection and consolidation method for experiment purposes.

It is also possible to conduct model comparisons to ensure that the best model have been chosen by referring to [32] and [33]. Besides, it is required to consider the advantages and disadvantages of the methodology that would yield the best possible return in money, time and effort. Lastly, it is required to anticipate and quantify the problems that may be encountered during the experiments and execution process.

### F. Phase 5: Design the Model

In this phase, several items need to be designed for the CLV prediction. First of all, it is required to design a comprehensive method for data collection, preprocessing, consolidation and reporting to handle both the structured and unstructured input data. Moreover, it must be able to start building the CLV predictive model with the details collected. The scope of the predictive model must be scalable according to the business needs. A dashboard can



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be used to display the CLV prediction output. Similarly, it is also required to design the experimental model and benchmarking values to evaluate the output. The predictive model needs to be revised regularly in order to increase its accuracy and performance.

### G. Phase 6: Conduct Testing

Testing is an important phase in a CLV prediction lifecycle. The team can set benchmarking values and models to compare the prediction outputs. One of the common benchmark values is called root mean square error (RMSE). The team can adjust the predictive variables accordingly if the RMSE value is high.

### H. Phase 7: Conduct Post Analysis

The post analysis phase is used to evaluate the model's performance. The performance can be measured from various aspects including data validity and integrity rate, business goals, Key Performance Indicators (KPI), processes functionality, return on investment (ROI) and net present value (NPV) of the customers.

The organization's top management must evaluate the model's performance and indicate whether it has met the vision and goals set in the earlier stage. The model is considered successful if it is able to produce accurate CLV of the customers in the respective companies.

### I. Phase 8: Record Documentation and Maintenance Work

The last phase of the iterative lifecycle is to record all the activities, resources and architecture of the CLV prediction lifecycle for future references. The documentation process will be much easier and less time consuming if the organizations use a project management tool to make recording and documentation starting from the early stage of the lifecycle. The important reports to be maintained are preliminary requirements report, analysis report, design report, full deployment report, user manuals, users' feedback reports and future work proposal.

The maintenance work can be done in several ways depending on the organization's timeline and budget. For example, minor research can be done to improve the CLV output display while major research such as dynamic CLV calculation and CLV break down values can be planned for future milestones. In short, it is required to evaluate the model regularly to ensure that it meets the business objectives and mission.

## IV. Discussion

In the proposed CLV prediction lifecycle, there are several inner iterative loops which revert phase 4 'Analyze the Requirements' and phase 5 'Design the Model' back to phase 3 'Plan the Process'. These inner iterative loops are incorporated to enable the users to go back to the previous phase when they are caught in some bottlenecks. The users must always be aware of the average time spent in each phase of the lifecycle. Good time management can save money and efforts. It is felt that more time should be allocated to vision definition, data investigation, analysis and preparation, model creation, as well as testing and validation. Similarly, more time should also be spent on regularly revising the predictive model.

## V. Future Work and Conclusion

Since CLV prediction research is getting mature, more sophisticated standards are required to further drive the research to a greater height. It is in our research pipeline to derive a more elaborated process template in the course of conducting the CLV predictions among the customers of a non-profit organization nurturing a sizable group of small and medium-sized enterprises. A research roadmap for incorporating the proposed prediction lifecycle and other critical issues will be further examined.

The proposed CLV prediction lifecycle has been designed as a guide for the CRM and CLV research communities for possible standardization in CLV research. This initial paper covers only a set of brief integrated guidelines that may help the interested parties to plan, analyze and design a CLV prediction process. It is hoped that the proposed lifecycle could eventually help to optimize the time, cost and resources for any organization engaging in CLV prediction activities.

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## References

[1] Torcy, Guy de, "Customer Lifetime Value", Available at http://www.cincom.com/common/pdf/Customer%20Lifetime%20Value%201.pdf, [Accessed on October 15, 2009]

[2] Svanholmer, B., "The Concept of CLV (Customer Lifetime Value)", Loyalty Mangement, http://www.loyaltymanagement.dk/pdf/theconceptofclv.pdf [Accessed on October 20, 2009]

[3] Taylor, J., "Optimizing Business Decisions – How Best to Apply Predictive Analytics", 2009, Decision Management Solutions, Webinar slides-9

[4] Hart, J., "Leveraging Predictive Analytics in Marketing Campaign", white paper, Hart Creative Marketing, Inc. Available at http://www.hartcreativemarketing.com/marketing-articles/leveraging-predictive-analytics.pdf [Accessed on October 20, 2009]

[5] Danylyszyn, A., "A Roadmap for Web Services-driven BPM - Learning through Example", SOA & WOA: Article, published on December 1, 2003

[6] Kelly, K., "Business Agility: Developing and Managing the BPM Roadmap", CIO Long Term Care Division, New York Life, June 23, 2009

[7] Olive, M., Rahmouni, H., Solomonides, T., "A European Healthgrid Roadmap", Share Project – Supporting and Structuring Healthgrid Activities & Research in Europe: Developing a Roadmap, October 2008



Authorized licensed use limited to: Cambridge Institute of Technology - Bengaluru. Downloaded on March 12,2026 at 09:23:39 UTC from IEEE Xplore. Restrictions apply.

[8] Bondavalli, A., Ceccarelli, A., Falai, L., Karlsson, J., Kocsis, I., Lollini, P., "Preliminary Research Roadmap", Project Report, Project funded by the European Community under the Seventh Framework Programme, September 30, 2008

[9] Nolte, T., Chhatwal, A., "Predictive analytics", LLC Workshop presentation, Availabe at: http://www.semilla-advisors.com/ Predictive%20Analytics%20(Platt's%20workshop%20031207).pdf

[10] Sloman, A., "What's a Research Roadmap for? Why do We Need One? How can We Produce One?" Research Roadmap Presentation: euCognition Meeting, Munich Jan 12 2007, Available at http://www.cs.bham.ac.uk/research/projects/cosy/presentations/ munich-roadmap-0701.pdf

[11] Michael P. Papazoglou, Paolo Traverso, Schahram Dustdar, Frank Leymann, Bernd J. Krämer, "Service-oriented Computing Research Roadmap", Dagstuhl Seminar Proceedings 05462 Service Oriented Computing (SOC), Available at: http://drops.dagstuhl.de/opus/volltexte/2006/ 524

[12] William M.K. Trochim, "Navigating the Knowledge Base, the Road Map" from Research Methods, Knowledge Base, Availabe at: http://www.socialresearchmethods.net/kb/ kbroad.php, [Accessed on September 6, 2009]

[13] Gillin P., (2007), "Choosing the Best CRM for Your Organization", white paper, Oracle Corporation and IT Business Edge

[14] Baohong Sun , Shibo Li , Catherine Zhou , ""Adaptive" Learning and "Proactive" Customer Relationship Management", (2006), Journal of Interactive Marketing, 82-96

[15] InvestorWords.com, http://www.investorwords.com/4238/ retention_rate.html , [Accessed on November 9, 2009]

[16] Investopedia dictionary, http://www.investopedia.com/terms/ r/returnoninvestment.asp, [Accessed on November 9, 2009]

[17] Lucas A., "Customer Lifetime Value Modelling", (2008), Rhino Risk Ltd, Conference Presentation Slides, Available at http://www.crc.man.ed.ac.uk/conference/archive/2008/presentations/alan-lucas.pdf , [Accessed on December 3, 2009]

[18] Allen B., Hamilton, "Customer Lifetime Value—Powerful Insights into a Company's Business and Activities", (2000), Insights, Communications Media & Technology Group, Vol 5, Issue 2.

[19] Sohrabi B., Khanlari A., (2007), "Customer Lifetime Value (CLV) Measurement Based on RFM Model", Iranian Accounting & Auditing Review, Spring 2007,Vol. 14 No. 47, pp 7- 20

[20] Hoekstra J., Huizingh E. ,(1999), "The Lifetime Value Concept in CustomerBased Marketing" , Journal of Market Focused Management, 3, 257274

[21] Shuang Zhang; Hongwu Guo; Congbao Qiu; Fengyan Hu; "Application of Customer's Lifetime Value Forecast Model to Classify Tobacco", Customers Control and Decision Conference, 2009. CCDC '09. Chinese 17-19 June 2009 Page(s):2512 - 2515

[22] Yun Chen, Huirong Zhang, Ping Zhu, "Study of Customer Lifetime Value Model Based on Survival-Analysis Methods", 2009,Computer Science and Information Engineering, WRI World Congress on Volume 7, March 31 2009-April 2 2009 Page(s):266 – 270

[23] Yi Wang, Sanguansintukul, S., Lursinsap, C., "The Customer Lifetime Value Prediction in Mobile Telecommunications", 2008,Management of Innovation and Technology, ICMIT 2008. 4th IEEE International Conference on 21-24 Sept. 2008 Page(s):565 – 569

[24] Nicolas Glady, Bart Baesens, Christophe Croux, "A Modified Pareto/NBD approach for Predicting Customer Lifetime Value", March 2009, Expert Systems with Applications: An International Journal, Volume 36 Issue 2, Publisher: Pergamon Press, Inc

[25] Chi-Wen Chen, Chyan Yang, Chiun-Sin Lin, "A Study of Discovering Customer Value for CRM: Integrating Customer Lifetime Value Analysis and Data Mining Techniques", Information and Management Science, Vol.2 No.1 , Page(s) 14-30

[26] Huan-Ming Chuang; Chia-Cheng Shen, "A Study on the Applications of Data Mining Techniques to Enhance Customer Lifetime Value — Based on the Department Store Industry", 2008, Machine Learning and Cybernetics, International Conference on, Volume 1, 12-15 July 2008 Page(s):168 – 173

[27] Annibal Parracho Sant'Anna, Rodrigo Otavio de Araujo Ribeiro, "Application of Stochastic Models to Determine Customer Lifetime Value for a brazilian Supermarket Network", Pesquisa Operacional, v.28, n.3, p.563-576, Setembro a Dezembro de 2008

[28] Paul D. Berger, NadaI . Nasr, "Customer Lifetime Value: Marketing Models and Applications", Journal of Interactive Marketing, Volume 12 / Number 1 / Winter 1998, page 17 - 30

[29] Predictive Analytics World Survey Report, Feb 2009, Available at http://www.predictiveanalyticsworld.com/Predictive-Analytics-World -Survey-Report-Feb-2009.pdf, [Accessed on September 10, 2009]

[30] "Defining the BPM vision: The critical first step to success", Archstone Research Paper, Archstone Consulting, 2007 Available at: http://www.archstoneconsulting.com/WorkArea/showcontent.aspx? id=1942

[31] Rong Liu, Jiayin Qi, "Dynamic Impact Factors of Customer Lifetime Value: An Empirical Study of M-zone", 2009, Eighth International Conference on Mobile Business, Pg 64 – 69

[32] Sunil Gupta et. al, 2006, "Modeling Customer Lifetime Value", Nov 2006, Journal of Service Research, vol. 9: pp. 139 - 155.

[33] Bas Donkers, Peter C. Verhoef, Martijn G. de Jong, "Modeling CLV: A Test of Competing Models in the Insurance Industry", Quant Market Econ (2007) 5:163−190



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